

Project Smart Trigger for ECE/CS 3992

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Abstract—Some people will read this and nothing else. What should they know about your proposal?

Index Terms—IEEE, IEEEtran, journal, L^AT_EX, paper, template.

I. PROBLEM STATEMENT

What is the problem the project is trying to solve?

II. SOLUTION SPACE

What are some approaches that people have taken to solve this problem? Sometimes an approach someone has taken then introduces new problems. So the description of the problem and solution take a problem -> solution -> problem -> solution -> ... format.

III. RELATED WORK

A more in-depth, formal exploration of the research and product development related to the problem space.

IV. SOLUTION / CONTRIBUTION

1-2 paragraphs summarizing your approach, what's exciting about it, why it will succeed

V. SAFETY, GLOBAL, SOCIETAL, ETHICAL CONSIDERATIONS

What are the public health, safety and welfare, global, cultural, societal, environmental, and economic factors related to the problem and solution space? What power does the proposed system provide, and to whom? What are the relevant professional ethical obligations to consider in the design of the system?

A. Factors Related to the Problem and Solution

1) *Public Health, Safety, and Welfare:* The main issue the proposed system addresses is the potential for negligent discharge of a firearm. This is because of the nature of guns in that they are designed to inflict harm on a target. The smart trigger would help to protect the public safety in a couple of ways. First the system would

be designed to prevent the gun from firing unless it is used by the owner. This would help to prevent accidental shootings by children or others that are not licensed to shoot a firearm. Secondly, if the gun was lost unless the module was removed the gun wouldn't be able to fire. This would help to prevent the gun from being used in crime or from being used in a negligent way. Lastly, the system is module meaning that if at any point it has a malfunction the module can be removed and the gun can be used as normal. This would help to prevent situations where a gun is rendered unusable due to malfunctions of the system. In the past since the mid 1990s, there have been numerous starts to create something similar to the smart trigger. These safety devices are known as Firearm Safety Devices (FSDs) and are most of the time mechanical as they are reliable and don't require a power source as shown in fig 1.

They also have some characteristics and features that are mentioned in [1]

- the option of integrating and combining mechanical devices with various authorization processes (token, password, biometric);
- hierarchical authorization;
- traceability and logging functions;
- networking and remote availability.



Fig. 1. Trigger and Cable Lock

- 2) *Global, and Cultural:*
- 3) *Societal and Economic:*
- 4) *Environmental:*

B. Power

VI. PROPOSED WORK

A project includes some things and excludes others you know you won't work on. Define more explicitly what is included and not included here. What will be demonstrated at project completion at Demo Day?

VII. PROJECT TASKS

Describe each task.

- "baseline deliverables" – things you will deliver
- "stretch goals" – things that you are planning to include, but the project will succeed even without them (and why)
- What components will be purchased and what will be built?
- What resources are needed to complete the tasks? These may be your own, or in a lab, or may be people who will help you.
- Task Interfaces: What are the important inputs and outputs of each individual component? Define them.
- Testing: How will each task be tested?
- Design approach: For each component, how will it be designed? What tools will be used (software, hardware, measurement, etc.)
- Current status: What tasks have been finished? What is left to be done?
- Schedule: A date when each task will be completed.
- Parts: Complete parts list with vendor name and information

VIII. INTEGRATION STRATEGY

How will the individual tasks/components be integrated? How will the integration be tested?

IX. RISK ASSESSMENT

What are the largest risks to successful demonstration of the project? Describe each risk as green, yellow, or red. At proposal stage, your goal is to have all risks in the "green" category, that is, you have de-risked any major risks to the project. Describe the de-risking process.

X. CONCLUSION

ACKNOWLEDGMENT

REFERENCES

- [1] E. S. Devices, "Personalized firearms and electronic safety devices," 2013.